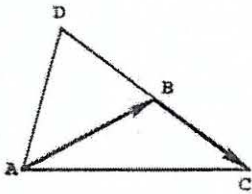


Vectors 1

1. In the figure B is the midpoint of CD . Name a vector that is equal to

a. $\vec{AB} + \vec{BC}$

b. $\vec{AB} - \vec{BC}$



2. If $\vec{AB} = \lambda \vec{CA}$, show that the points A , B and C are collinear.

3. Find the value of λ for which the points P and Q are coincident, given that $\vec{PB} + 2\vec{PQ} = (\lambda - 2)\vec{QB}$

4. Express each of the following vectors in the form $\lambda \vec{a} + \mu \vec{b}$:

a. $\vec{OA}$

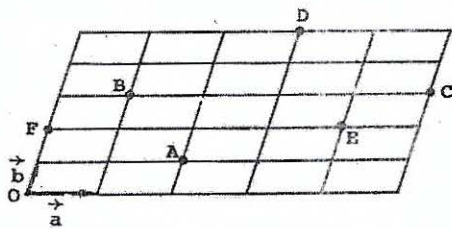
c. $\vec{BA}$

e. $\vec{EB}$

b. $\vec{OC}$

d. $\vec{DA}$

f. $\vec{FC}$



5. Express each of the following as a single vector

a. $\vec{OA} + \vec{AB} + \vec{BC} + \vec{CD} + \vec{DE}$

c. $\vec{XY} + \vec{YA} - \vec{ZY} + \vec{ZX}$

b. $\vec{AB} - \vec{DC} + \vec{DA}$

6. If C is the midpoint of AB and R is the midpoint of PQ , show that $2\vec{RC} = \vec{PA} + \vec{QB}$

7. Find λ if C and A are coincident, and $\vec{CB} + \vec{CA} = (\lambda - 1)\vec{AB}$

8. If D , E and F bisect AB , BC and CA show that

a. $\vec{AF} + \vec{CE} = \vec{AD}$

b. $\vec{PD} + \vec{PE} + \vec{PF} = \vec{PA} + \vec{PB} + \vec{PC}$ where P is any point

c. $\vec{AB} + \vec{AC} = 2\vec{AE}$

d. Show also that $\vec{DF}$ is parallel to $\vec{BC}$ and is equal to half of it

9. If $ABCDEF$ are the vertices of a regular hexagon, show that $\vec{AB} + \vec{AC} + \vec{AD} + \vec{AE} + \vec{AF} = 3\vec{AD}$

10. Express each of the following vectors in the form $\lambda \vec{a} + \mu \vec{b}$:

a. $\vec{OA}$

f. $\vec{BE}$

k. $\vec{BD} + \vec{BG}$

b. $\vec{OC}$

g. $\vec{CG}$

l. $\vec{EA} - \vec{EC}$

c. $\vec{BA}$

h. $\vec{FG}$

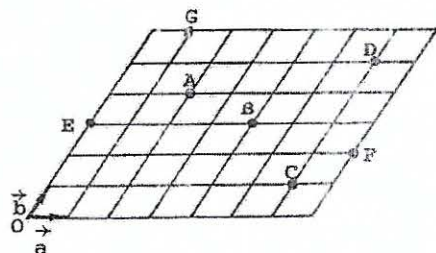
m. $\vec{OA} + \vec{OB} + \vec{OC}$

d. $\vec{AB}$

i. $\vec{OA} + \vec{OG}$

e. $\vec{AD}$

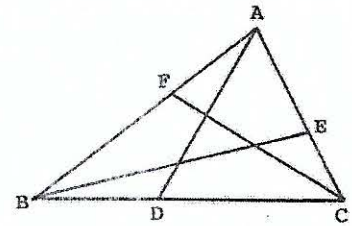
j. $\vec{OA} + \vec{OF}$



11. If $\vec{a}$, $\vec{b}$ are the vectors $\vec{AB}$, $\vec{BC}$ determined by two adjacent sides of a regular hexagon, what are the vectors determined by the other sides taken in order?

12. If O, A, B, C are points such that $\ell\overrightarrow{OA} + m\overrightarrow{OB} + n\overrightarrow{OC} = \vec{0}$ where $\ell + m + n = 0$, prove that C is the point dividing AB in the ratio $m : \ell$.

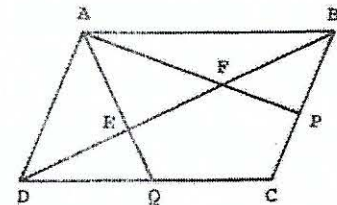
13. In a triangle ABC , the points D, E, F divide the sides BC, CA, AB internally in the ratio $k : 1$. Prove that $\overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF} = \vec{0}$



14. The points D, E, F divide the sides BC, CA, AB of a triangle in the ratio $1 : 4, 3 : 2$ and $3 : 7$ respectively.

Show that the sum of the vectors $\overrightarrow{AD}, \overrightarrow{BE}, \overrightarrow{CF}$ is parallel to $\overrightarrow{CK}$, where K divides AB in the ratio $1 : 3$.

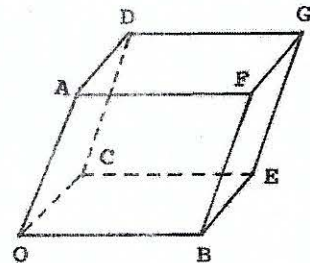
15. In the adjoining figure, $ABCD$ is a parallelogram with P and Q the midpoints of sides BC and CD respectively. Prove that AP and AQ trisect diagonal BD at the point E and F .



16. P is any point in the plane of a triangle ABC . Show that $\overrightarrow{PA} + \overrightarrow{PB} + \overrightarrow{PC} = 3\overrightarrow{PG}$ where G is the centroid of the triangle ABC .

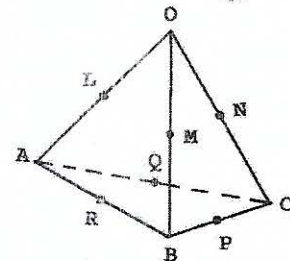
17. The corners A, B, C of the parallelepiped have position vectors $\vec{a}, \vec{b}, \vec{c}$ from the corner O . Find in terms of $\vec{a}, \vec{b}, \vec{c}$:

- The position vectors of D, E, F, G
- $\overrightarrow{AC}, \overrightarrow{FE}$
- The position vectors of the midpoints of AC and EF
- The position vectors of the midpoints of OG, AE, BD, CF . What do you deduce?



18. The corner A, B, C of a tetrahedron have position vectors $\vec{a}, \vec{b}, \vec{c}$ from the centre O . Find the position vectors of

- The midpoints of OA, OB, OC (L, M, N)
- The midpoints of BC, CA, AB (P, Q, R)
- The midpoints of LP, MQ, NR . What do you deduce?



19. a. Three ships are equidistant from a lighthouse. Find the position vector of the lighthouse if the position vectors of the ships are $3\vec{i} + \vec{j}, 2\vec{i} - 4\vec{j}$ and $\vec{i} - 3\vec{j}$ and distance is measured in km .

b. Use vector to show that the perpendicular bisectors of the side of a triangle are concurrent.

Answer:

- a. $\overrightarrow{AC}$; b. $\overrightarrow{AD}$
- $\lambda = 3$
- a. $2\vec{a} + \vec{b}$; b. $5\vec{a} + 3\vec{b}$; c. $\vec{a} - 2\vec{b}$; d. $-\vec{a} - 4\vec{b}$; e. $-3\vec{a} + \vec{b}$; f. $5\vec{a} + \vec{b}$
- a. $\overrightarrow{OE}$, b. $\overrightarrow{CB}$, c. $\overrightarrow{YA}$
- 2
- a. $2\vec{a} + 4\vec{b}$; b. $6\vec{a} + \vec{b}$; c. $-2\vec{a} + \vec{b}$
- d. $2\vec{a} - \vec{b}$; e. $4\vec{a} + \vec{b}$; f. $-4\vec{a}$; g. $-5\vec{a} + 5\vec{b}$; h. $-6\vec{a} + 4\vec{b}$; i. $3\vec{a} + 10\vec{b}$; j. $9\vec{a} + 6\vec{b}$; k. $-\vec{a} + 5\vec{b}$; l. $-4\vec{a} + 3\vec{b}$; m. $12\vec{a} + 8\vec{b}$
- $\vec{b} - \vec{a}, -\vec{a}, -\vec{b}, \vec{a} - \vec{b}$
- a. $\vec{a} + \vec{c}, \vec{b} + \vec{c}, \vec{a} + \vec{b}, \vec{a} + \vec{b} + \vec{c}$
- b. $\vec{c} - \vec{a}, \vec{c} - \vec{a}$; c. $\frac{1}{2}(\vec{a} + \vec{c})$, $\frac{1}{2}\vec{a} + \vec{b} + \frac{1}{2}\vec{c}$; d. $\frac{1}{2}(\vec{a} + \vec{b} + \vec{c})$
- a. $\frac{1}{2}\vec{a}, \frac{1}{2}\vec{b}, \frac{1}{2}\vec{c}$; b. $\frac{1}{2}(\vec{b} + \vec{c}), \frac{1}{2}(\vec{c} + \vec{a}), \frac{1}{2}(\vec{a} + \vec{b})$; c. $\frac{1}{4}(\vec{a} + \vec{b} + \vec{c})$
- a. $\frac{5}{3}(2\vec{i} - \vec{j})$